



UNIVERSITY MANAGEMENT SYSTEM DOCUMENTATION

1.Introduction

1.1Purpose

The purpose of this project is to practice Object-Oriented Programming concepts by building a simulated University Management System that manages students, courses, instructors, and registration operations.

1.2 Objectives

- Manage student records.
- Manage courses and instructors.
- Handle course registration.
- Organize departments.
- Apply OOP concepts in a real-world system.

1.3 Scope

The system allows administrators to add, update, delete, and display university data including students, instructors, departments, and courses.

2. System Overview

The system simulates how universities organize academic information and student enrollment operations digitally instead of manual paperwork

3. Functional Requirements

Student Management

- Add student
- Remove student
- Update student information
- Display student details
- Search student by ID

Course Management

- Add courses
- Update course information
- Display available courses
- Search courses by ID
- Check for duplicate courses

Registration System

- Register students in courses
- Drop courses
- Prevent duplicate registration
- Check course capacity before registration
- Check prerequisite courses before registration
- Validate enrolled credit hours per student

Academic Evaluation

- Calculate GPA for students
- Determine Honor Roll students

Instructor Management

- Assign instructors to courses
- Remove instructors from courses
- View instructor assigned courses

4. Non-Functional Requirements

The system should provide clear menus and organized outputs for easier navigation.

5. Object-Oriented Programming Concepts Used

Encapsulation

Private attributes with public getters and setters.

Polymorphism

Function overriding for displaying user information.

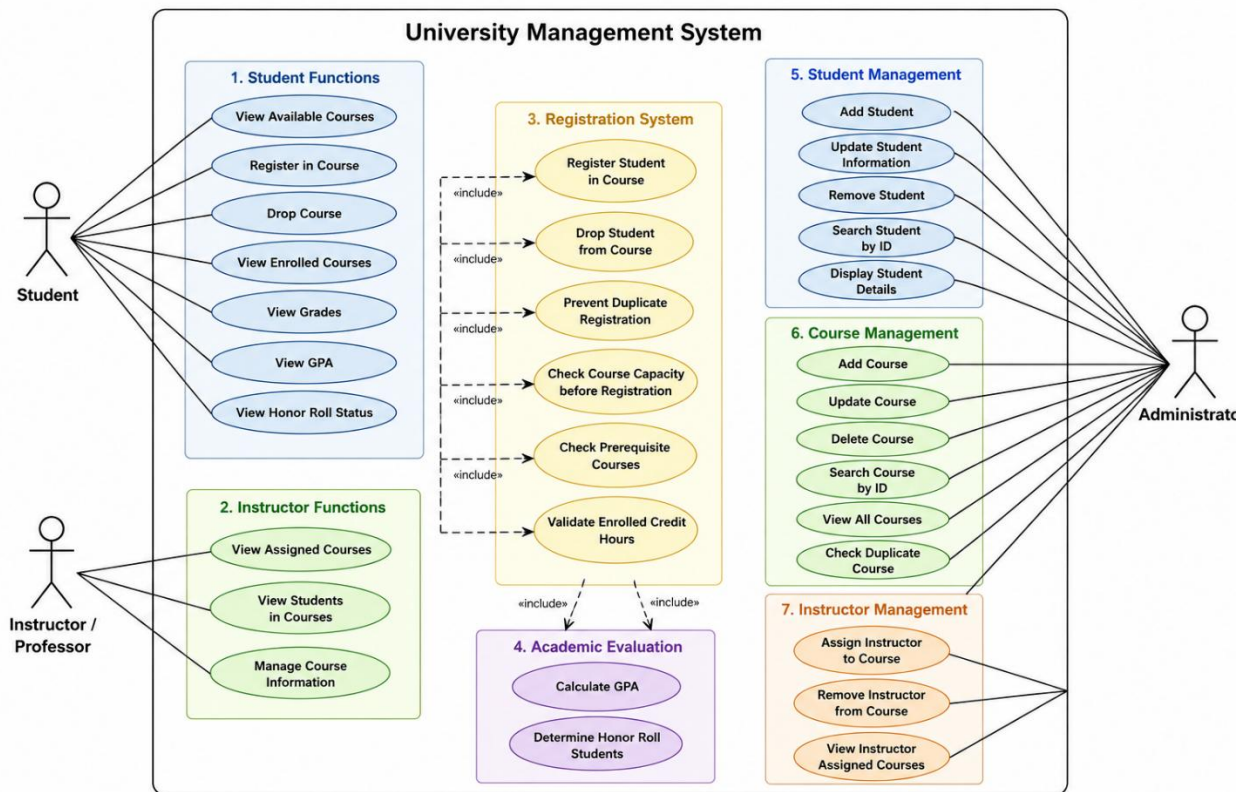
Abstraction

Using abstract classes/interfaces where needed.

6. System Design

Class	Responsibility
Student	Store student information
Course	Store course details
Department	Organize departments
Registration	Handle enrollment operations

7. UML Diagrams



8. Algorithms and Logic

- Student registration validation
- Searching for courses
- Duplicate checking

9. Challenges Faced

- Managing relationships between classes
- Avoiding duplicate enrollments
- File organization
- Debugging getter/setter issues

10. Future Improvements

- GUI interface
- Database integration
- Authentication system
- Online registration

11. Conclusion

This project provided practical experience in applying OOP principles to build a structured management system that simulates real university operations.